Causal-Factored Cross-City Model Transfer for Bus Holding Control with Open Transit Data

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Abstract

Hybrid offline-to-online reinforcement learning for transit control is attractive because operators often have rich historical data but only imperfect simulators. However, a bus holding policy tuned in one city may fail in another city unless the target simulator is manually calibrated, and such calibration is expensive. We study this cross-city transfer problem for bus holding using four open transit city bundles: Singapore, Austin/CapMetro, Halifax Transit, and MBTA. We propose Causal-Factored Cross-City Model Transfer (CFCMT), a mechanism-wise residual world-model adapter that decomposes dense simulator corrections into headway, demand/dwell, speed, and reward mechanisms, and optionally reweights source cities by static speed, headway, route, and demand similarity. Against a dense H2O^{+}-style residual baseline, CFCMTis evaluated under a strict leave-one-city-out protocol in which each of the four cities is the target exactly once. On static-derived cross-city dynamics validation, unweighted CFCMTbeats H2O^{+}on all four targets with mean total-MSE ratio 0.759, while similarity-weighted CFCMTalso wins on all four targets and lowers the mean ratio to 0.578. In one-step policy validation, CFCMTimproves reward/regret on three of four targets and lowers bunching rate on all four targets. Calibration-budget, per-mechanism, ablation, route-bootstrap, source-subset, and generator-bias checks support the same main trend while showing that the advantage depends on multi-source diversity and should not be overstated as superiority on every small city subset. The resulting contribution is a reproducible, reviewer-facing protocol and an empirically supported boundary condition: under open static-data-derived validation with little or no target-city calibration, mechanism-factored transfer generalizes better than dense residual correction, but live AVL/APC trajectory validation remains necessary before operational claims.

1 Introduction

Bus holding control aims to reduce headway instability by delaying selected vehicles at stops. The control problem is local in execution but system-wide in consequence: a hold action changes forward and backward headways, passenger waiting, dwell, and downstream bunching. Modern reinforcement learning (RL) methods can learn holding policies in simulation, but practical deployment is constrained by two persistent gaps. First, high-fidelity simulation and manual calibration are expensive to reproduce in each city. Second, public data availability differs sharply across agencies: one city may expose speed bands and passenger OD data, while another may only provide GTFS schedules and coarse ridership summaries.

Hybrid offline-to-online methods such as H2O^{+}learn from both logged data and simulator interaction by correcting or reweighting dynamics gaps (Niu et al., 2022, 2025). This is a natural starting point for transit systems, where a historical buffer and a lower-fidelity simulator are often available. The difficulty is cross-city transfer. A dense monolithic residual model can fit simulator error in the source cities, but its parameters mix mechanisms that may shift differently across cities: schedule headways, route speeds, stop demand, and dwell behavior do not all change together. If cross-city transfer depends on localized changes in mechanisms, a factored representation should be more stable than a dense one (Bengio et al., 2020; Feng et al., 2022).

This paper studies that hypothesis in a deliberately conservative setting. We do not claim that the current data are vehicle-level real-world trajectories. Instead, we construct full-route open transit city bundles and evaluate models on static-data-derived offline dynamics targets. The protocol is meant to answer a narrower question: when only open static data and an uncalibrated simulator-like transition generator are available, does a mechanism-factored residual adapter transfer better across cities than a dense H2O^{+}-style residual correction?

We introduce Causal-Factored Cross-City Model Transfer (CFCMT). CFCMTfits residual corrections at the level of output mechanisms rather than one dense residual over all outputs. For each target quantity, it uses a mechanism-specific sparse parent set: headway and gap outputs depend primarily on neighboring headways, target headway, speed, time, and action; demand and dwell outputs depend on waiting passengers, speed, target headway, time, and action; speed outputs use speed, time, and route-position features; reward uses policy-relevant headway and waiting quantities. We also add a city-similarity source weighting rule that uses only static descriptors available before target calibration.

The paper contributes:

1. A cross-city open-transit validation protocol using four full-route city bundles and a strict leave-one-city-out evaluation unit.
2. A mechanism-wise residual transfer model and a static descriptor based source weighting rule.
3. A set of reviewer-facing diagnostics: source-weight sensitivity, source-subset robustness, generator-bias/noise robustness, and traffic-domain policy metrics.
4. An explicit boundary on the claim: the evidence supports static-derived cross-city transfer and not yet operational deployment without vehicle-level validation.

2 Related Work

Bus holding control. Classical holding policies analyze headway stability and bunching through control rules and queueing approximations (Daganzo, 2009; Xuan et al., 2011). Recent work has applied deep RL and multi-agent RL to bus holding under stochastic demand and travel times (Alesiani and Gkiotsalitis, 2018; Wang and Sun, 2020; Liu et al., 2023; Wang and Sun, 2023). These studies typically focus on one calibrated network or one simulation benchmark. Our focus is different: we evaluate whether a learned dynamics correction transfers across city bundles with heterogeneous open data sources.

Hybrid offline-to-online RL and dynamics gaps. H2O^{+}and H2O^{+}-style methods address the mismatch between offline data and online simulator interaction by estimating dynamics-related weights or residuals (Niu et al., 2022, 2025). Related sim-to-real and offline-to-online methods use domain classifiers, calibrated Q-learning, residual dynamics, or world-model adaptation to reduce transfer error (Eysenbach et al., 2021; Nakamoto et al., 2023; Lanier et al., 2025). In this paper, H2O^{+}is treated as the dense residual-correction baseline: it can learn flexible source-city corrections, but the correction is not explicitly decomposed by transit mechanism.

Causal and factored transfer. The independent causal mechanisms principle suggests that distribution shifts may be localized when represented in appropriate mechanisms (Bengio et al., 2020). Factored adaptation methods use this idea to improve non-stationary RL and transfer by separating latent change factors (Feng et al., 2022). CFCMTapplies this principle to bus holding dynamics: rather than learning one dense correction, it learns mechanism-wise residuals aligned with interpretable transit outputs. The approach is intentionally lightweight in this paper: the reported cross-city experiments use sufficient-statistics ridge models for reproducibility, while the repository also contains neural causal-factored residual modules for later end-to-end training.

Traffic simulation and open transit data. SUMO provides a standard microscopic traffic simulation platform (Lopez et al., 2018). For cross-city public transit studies, however, fully calibrated SUMO networks and vehicle-level AVL/APC data are rarely available across many agencies. We therefore use open static data to create full-route city bundles. This makes the experiments reproducible but also limits their interpretation; the static-derived targets are not a substitute for live operational trajectory validation.

3 Problem Setting

Let $s_t \in \mathcal{S}$ denote a bus-system state at a decision event and $a_t \in \mathcal{A}$ denote a holding action in seconds. The state contains line and station identifiers, time of day, direction, forward and backward headways, waiting passengers, target headway, base dwell, co-line headways, gap, and segment speed. The transition target is

$$\mathbf{y}_t = (h^+_{t+1}, h^-_{t+1}, q_{t+1}, d_{t+1}, g_{t+1}, \bar{h}^+_{t+1}, \bar{h}^-_{t+1}, v_{t+1}, r_{t+1}),$$

covering next headways, waiting passengers, dwell, gap, co-line headways, speed, and reward.

Each city c provides a bundle $\mathcal{D}_c$ of route, timetable, speed, and demand tables. From these tables we construct paired simulator and static-derived target transitions:

$$\mathbf{y}^{\text{sim}} = f_0(s, a), \qquad \mathbf{y}^{\text{target}} = f_c(s, a),$$

and train residual models on

$$\Delta_c(s, a) = \mathbf{y}^{\text{target}} - \mathbf{y}^{\text{sim}}.$$

For cross-city validation, source cities $\mathcal{C}_{\text{src}}$ are used for fitting and a held-out target city $c^\star$ is used only for evaluation.

The main evaluation unit is a city target, not a route. For four cities, the strict leave-one-city-out protocol has exactly four target evaluations:

$$\{(\mathcal{C} \setminus \{c\}) \to c : c \in \mathcal{C}\}.$$

This avoids counting repeated target cities as independent evidence.

Claim scope. The target transitions are static-data-derived pseudo-observations. They combine agency data, schedule-derived speeds or headways, passenger-demand summaries, and deterministic transition rules. They are useful for reproducible cross-city validation, but they are not vehicle-level AVL/APC ground truth. Accordingly, the paper evaluates transfer robustness under open static-data-derived dynamics, not operational field performance.

4 Method

4.1 Dense H2O$^+$-style residual baseline

The H2O$^+$-style baseline fits one dense residual model over all transition outputs:

$$\hat{\Delta}_{\text{dense}}(s,a) = \phi_{\text{dense}}(s,a)^\top B,$$

where ϕ_{dense} includes normalized state/action features, squares, action interactions, and selected pairwise products. At evaluation time the corrected prediction is

$$\hat{\mathbf{y}}_{\text{H2O}^+}(s,a) = f_0(s,a) + \hat{\Delta}_{\text{dense}}(s,a).$$

The baseline is intentionally strong: it can use many cross terms and is fit on all source-city transitions.

4.2 Causal-factored residual transfer

CFCMTdecomposes the residual by output mechanism:

$$\hat{\Delta}_j(s,a) = \phi_j(s,a)^\top \beta_j, \qquad j \in \{1,\ldots,9\}.$$

The feature map ϕ_j is chosen according to the output mechanism. Headway and gap residuals use current headways, target headway, speed, action, and time. Waiting and dwell residuals use waiting passengers, target headway, speed, station fraction, action, and time. Speed residuals use current speed, event time, and station position. Reward residuals use policy-relevant headway deviations, waiting passengers, gap, speed, action, and time. The corrected prediction is

$$\hat{\mathbf{y}}_{\text{CFCMT}}(s,a) = f_0(s,a) + \left(\hat{\Delta}_1(s,a),\ldots,\hat{\Delta}_9(s,a)\right).$$

All reported models use ridge-regularized sufficient statistics, so the experiment can be reproduced without stochastic neural training.

4.3 Source-city similarity weighting

For each city we compute a static descriptor z_c from speed quantiles, headway quantiles, route-distance quantiles, peak-demand hour, and the 24-hour demand-share vector. For a target city t and source city c, the distance is computed after feature-wise standardization over the source-plus-target set:

$$d(c,t) = \frac{1}{\sqrt{m}}\left\|\frac{z_c-\mu}{\sigma} - \frac{z_t-\mu}{\sigma}\right\|_2.$$

The source weight is

$$w_c = \frac{\exp(-d(c,t)/T) + \lambda \max_{c'} \exp(-d(c',t)/T)}{\sum_{k\in\mathcal{C}_{\text{src}}}\left[\exp(-d(k,t)/T) + \lambda \max_{c'} \exp(-d(c',t)/T)\right]}.$$

The default values are $T = 1.0$ and $\lambda = 0.05$. The source sufficient statistics are rescaled so that these weights control the effective number of source transitions rather than simply reflecting raw city size.

city	env_key	demand_evidence	traffic_evidence	lines	stops	segments	timetables	transitions	median_route_km	median_speed_mps	median_headway_s	peak_demand_hour	demand_source	traffic_source
Singapore	singapore_lta_all	observed stop OD/PV monthly	observed speed bands	789	25873	25906	56545	1243488	17.0	9.58	690.0	7	LTA DataMall PV/ODBus 202604, WEEKDAY monthly totals divided to typical weekday	LTA TrafficSpeedBands snapshot nearest-matched to stop segments
Austin / CapMetro	austin_capmetro_all	schedule proxy	schedule derived	234	6464	6464	28329	310272	16.1678	6.19	1200.0	14	GTFS schedule headway proxy OD; no public no-key stop-level ridership found	GTFS schedule-derived segment speed
Halifax Transit	halifax_transit_all	route APC apportioned	schedule derived	176	5551	5551	8710	266448	12.3693	6.84	1140.0	15	Halifax route-level half-hour APC apportioned across GTFS route-direction-patterns	GTFS schedule-derived segment speed
MBTA	mbta_all	stop board/alight apportioned	schedule derived	940	20195	20195	67091	969360	7.6405	4.01	600.0	7	MBTA Fall 2025 stop-level boardings/alightings apportioned across GTFS route-direction-patterns	GTFS schedule-derived segment speed

Table 1: Open transit city bundles. Evidence levels distinguish observed passenger/traffic data from schedule-derived proxies.

4.4 One-step policy validation

For policy-level validation, each method evaluates a discrete hold set $\{0, 15, 30, 45, 60\}$ seconds under a set of headway-stress states. The policy selects the action with the highest predicted reward under its corrected transition model. The selected action is then scored against the static-derived target transition. We report reward, regret to an oracle action, mean absolute headway error, bunching rate, large-gap rate, and mean hold seconds. This is a policy-level one-step lookahead benchmark, not a live SUMO rollout.

5 Experiments

5.1 Data

We evaluate on four full-route city bundles. Singapore uses LTA DataMall passenger-volume and OD bus data with traffic speed bands. Austin/CapMetro uses GTFS schedules with demand proxied from scheduled headways. Halifax Transit uses route-level APC summaries apportioned over GTFS route-direction patterns. MBTA uses Fall 2025 stop-level boardings and alightings apportioned over GTFS route-direction patterns. Table 1 summarizes the bundles and their evidence levels.

5.2 Validation protocols

Primary dynamics validation. The main dynamics experiment is strict leave-one-city-out. Each city is held out once as the target, and all remaining cities are used as sources. The metric is total MSE averaged over the nine transition outputs. Ratios below 1 indicate improvement over the dense H2O^{+}-style residual baseline.

Policy validation. The one-step policy experiment uses the same source-target splits and evaluates action selection under headway-stress states. We report reward/regret and traffic-domain metrics. Bunching rate is the fraction of decisions in which either adjacent headway is below 0.5 times the target headway. Large-gap rate is the fraction in which either adjacent headway exceeds 1.5 times the target headway.

Reviewer-facing robustness checks. We include diagnostics targeted at common review objections. First, a target-construction audit separates observed, apportioned, schedule-proxy, and deterministic target components. Second, a target-route calibration-budget sweep adds 0%, 1%, 5%, 10%, 25%, or 100% of held-out target-city routes to the source sufficient statistics and evaluates on the remaining target routes; the 100% point is an in-sample oracle reference. Third, per-mechanism errors decompose total MSE into headway/gap, demand/dwell/reward, and speed groups. Fourth, mechanism ablations compare the proposed parent sets against wrong mechanism groupings, action/time-only features, and dense matched-capacity variants. Fifth, source-weight sensitivity varies $T \in \{0.5, 1.0, 2.0\}$ and $\lambda \in \{0, 0.05, 0.1\}$, while source-subset and pairwise-source checks remove major source families. Sixth, generator robustness perturbs output groups or adds measurement-noise terms to the evaluation sufficient statistics.

experiment	paper_role	unit	scope_note
strict_leave_one_city_out	primary cross-city dynamics evidence	city target	four unique targets; no duplicated target counted as independent
generator_robustness	bias/noise defense	scenario x city target	static-derived target perturbation; not real AVL/APC trajectory ground truth
source_subset_robustness	data-source confound defense	city subset	reports excluding Singapore and excluding Austin proxy-demand subsets
cross_city_policy_validation	primary one-step policy evidence	city target	one-step lookahead, not live SUMO rollout
sampled_rollout	auxiliary sanity check only	BusSimEnv episode	50 policy episodes across runnable line envs; do not use as primary claim

Table 2: Experimental scope and paper role. The primary evidence is strict city-level leave-one-out dynamics and one-step policy validation.

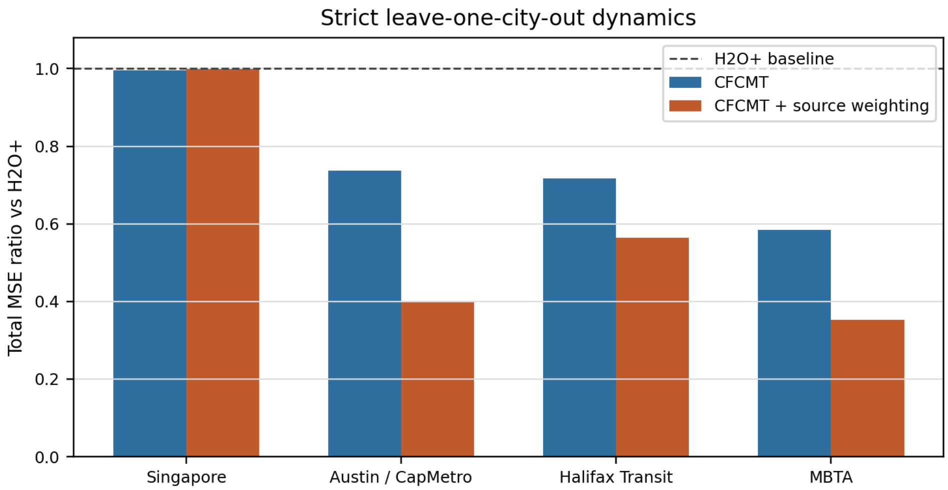

Figure 1: Strict leave-one-city-out dynamics validation. Bars show total-MSE ratio relative to $H2O^+$. The dashed line is parity with $H2O^+$.

Auxiliary executable rollout. Sampled BusSimEnv rollout is reported only as an auxiliary sanity check because only Austin and Halifax have runnable generated line environments. For this rollout, each policy is evaluated on the same line-level random seed, including Python signal-delay randomness, and live observations are mapped back to the static-transition feature schema before model scoring. The policy score uses the BusSimEnv linear headway reward rather than the static one-step reward.

6 Results

6.1 Strict leave-one-city-out dynamics

Figure 1 and table 3 show the primary dynamics result. Unweighted CFCMTbeats the dense $H2O^+$-style residual baseline on all four held-out cities, with mean total-MSE ratio 0.759. Similarity-weighted CFCMTalso wins on all four targets and improves the mean ratio to 0.578. The largest gains occur for Austin, Halifax, and MBTA. Singapore is the hardest target: unweighted CFCMTis only slightly better than $H2O^+$, and similarity weighting is slightly worse than unweighted CFCMTwhile still remaining below the $H2O^+$baseline.

6.2 Policy-level validation

The strict one-step policy summary is more mixed but still favorable. CFCMTimproves reward/regret on three of four targets, improves mean absolute headway error on three of four,

target_city	source_cities	target_transitions	h2oplus_total_mse	cfcmt_total_mse	weighted_cfcmt_total_mse	cfcmt_vs_h2oplus_ratio	weighted_cfcmt_vs_h2oplus_ratio	weighted_cfcmt_vs_unweighted_ratio
Singapore	austin_capmetro_all, halifax_transit_all, mbta_all	1243488	2254.8548	2246.6693	2251.7081	0.9964	0.9986	1.0022
Austin / CapMetro	singapore_lta_all, halifax_transit_all, mbta_all	310272	91.9724	67.8254	36.6531	0.7375	0.3985	0.5404
Halifax Transit	singapore_lta_all, austin_capmetro_all, mbta_all	266448	97.1175	69.5992	54.7536	0.7166	0.5638	0.7867
MBTA	singapore_lta_all, austin_capmetro_all, halifax_transit_all	969360	318.9925	186.4177	112.2698	0.5844	0.352	0.6022

Table 3: Strict leave-one-city-out dynamics results. Ratios below 1 indicate lower total MSE than $H2O^+$.

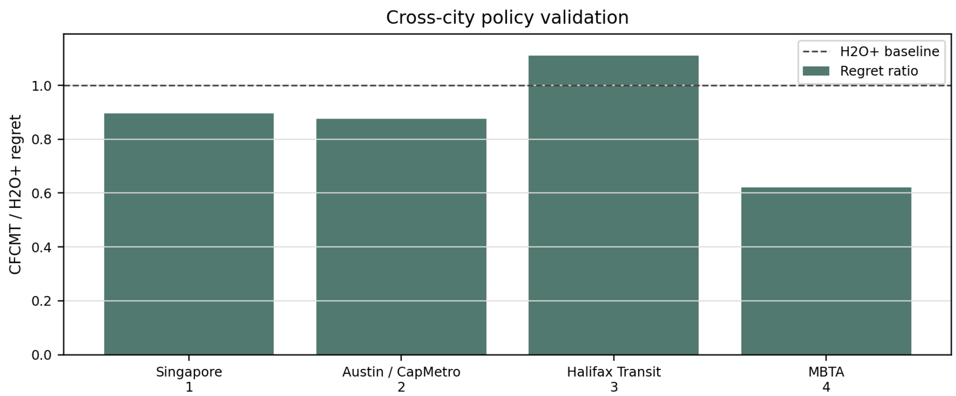

Figure 2: Strict leave-one-city-out one-step policy regret ratio. Ratios below 1 favor CFCMTover $H2O^+$.

and lowers bunching rate on all four targets. The mean regret ratio is 0.876, and the mean bunching-rate ratio is 0.644. Halifax is the main policy exception: CFCMTlowers headway error slightly but has worse reward/regret than $H2O^+$. This is consistent with a limited policy claim: mechanism-factored transfer improves cross-city dynamics strongly and often improves one-step policy quality, but it is not a universal policy dominance result.

6.3 Source weighting is stable under a small grid

The default source weighting uses $T = 1.0$ and $\lambda = 0.05$. Across the 3×3 grid, all nine settings beat $H2O^+$on all six configured cross-city splits. The mean weighted-CFCMT/$H2O^+$total-MSE ratio ranges only from 0.6025 to 0.6149. The best setting in the grid is $T = 0.5, \lambda = 0$, but the default is not selected by post-hoc optimization and remains within the same narrow band.

6.4 Calibration budget and mechanism-level errors

The target-route calibration-budget sweep tests whether a calibrated dense residual baseline quickly erases the zero-calibration CFCMTadvantage. At 0% target calibration, weighted CFCMThas mean ratio 0.578 against the dense source-only baseline. When 1%, 5%, 10%, and 25% of target routes are added to the dense baseline, the uncalibrated weighted-CFCMT/H2O ratio remains below 1 on three of four strict targets, with mean ratios 0.581, 0.586, 0.617, and 0.677, respectively. The 100% target-route point is an in-sample oracle reference and should not be interpreted as deployable zero-calibration evidence.

Per-mechanism errors show that the total-MSE advantage is not driven by one output group only. Weighted CFCMTimproves all three mechanism groups on average: headway/gap ratio 0.574, demand/dwell/reward ratio 0.771, and speed ratio 0.668. Headway/gap is the largest contributor to the main gain, which is consistent with the holding-control objective.

split	target_city	h2oplus_reward	cfcmt_reward	cfcmt_reward_gain	cfcmt_regret_ratio	cfcmt_headway_error_ratio	h2oplus_bunching_rate	cfcmt_bunching_rate	cfcmt_bunching_rate_ratio	h2oplus_large_gap_rate	cfcmt_large_gap_rate	cfcmt_large_gap_rate_ratio	cfcmt_hold_seconds_delta
leave_one_city_out_all::singapore_lta_all	Singapore	-0.3338	-0.3337	0.0001	0.8959	1.0007	0.0024	0.0024	0.9991	0.0016	0.0016	1.0	-0.3597
leave_one_city_out_all::austin_capmetro_all	Austin / CapMetro	-0.3326	-0.332	0.0007	0.875	0.9995	0.0	0.0	0.0	0.0001	0.0	0.0	0.3909
leave_one_city_out_all::halifax_transit_all	Halifax Transit	-0.3307	-0.3312	-0.0005	1.1113	0.9998	0.0	0.0	0.9355	0.0	0.0	1.0	1.4814
leave_one_city_out_all::mbta_all	MBTA	-0.3429	-0.338	0.0049	0.6201	0.9967	0.0006	0.0004	0.6396	0.0008	0.0002	0.3174	-1.1321

Table 4: Strict leave-one-city-out one-step policy metrics. The table includes reward, regret ratio, headway error ratio, bunching, large-gap rate, and hold-time difference.

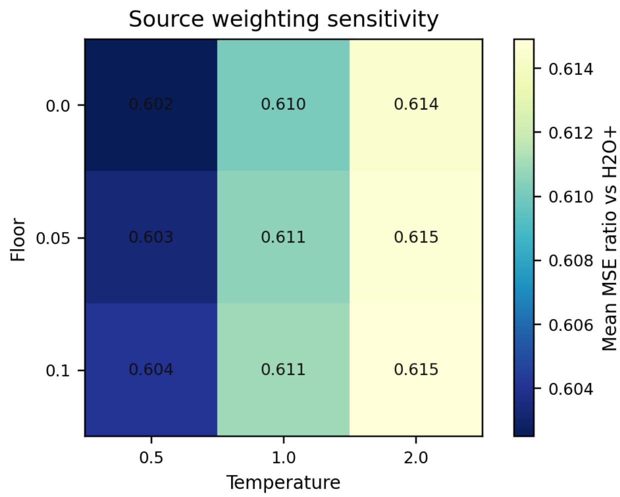

Figure 3: Source-weighting sensitivity. Each cell reports mean total-MSE ratio versus H2O$^+$across configured cross-city splits.

6.5 Robustness and boundary conditions

Generator robustness supports the main trend but also identifies failure modes. Weighted CFCMT-beats H2O$^+$on every city target in five of seven perturbation/noise scenarios. Under headway-gap bias and mixed-mechanism bias, the result weakens to three of four targets. Thus the evidence does not say that CFCMTis invariant to any target-generation change; it says that the result is robust to several plausible perturbations.

Mechanism ablations support the mechanism-specific interpretation, with a caveat. Wrong mechanism groupings and action/time-only features are much worse than H2O$^+$, with mean ratios above 7. Dense matched sparse features are close to full CFCMT, with mean ratio 0.772, which means the result should be described as evidence for mechanism-wise sparse transfer rather than definitive causal identification.

The source-subset check is the strongest limitation. With all four cities, weighted CFCMThas mean ratio 0.578. Excluding Austin remains favorable with mean ratio 0.663. Excluding Singapore makes the result nearly tied with H2O$^+$, with mean ratio 0.998. The two-city Halifax-MBTA pair is slightly worse than H2O$^+$, with ratio 1.042. This indicates that source diversity matters: the method should be framed as multi-source cross-city transfer, not as guaranteed transfer between every pair of cities.

Route-level bootstrap confidence intervals reinforce the same boundary. Unweighted CFCMThas a route-bootstrap ratio confidence interval below 1 for all four targets. Weighted CFCMThas intervals below 1 for Austin, Halifax, and MBTA, but Singapore crosses parity with interval $[0.993, 1.002]$. The conservative reading is that source weighting improves the mean strongly, while Singapore remains the least secure target.

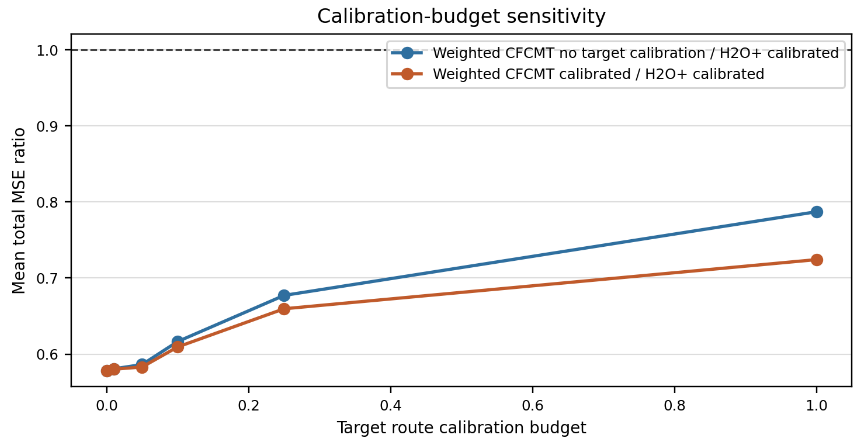

Figure 4: Target-route calibration-budget sensitivity. Ratios compare weighted CFCMTwith no target calibration or the same target budget against a dense H2O^{+}-style baseline using that target budget.

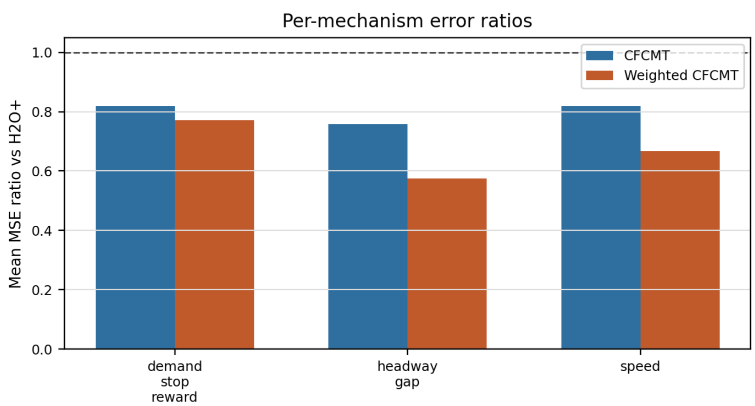

Figure 5: Per-mechanism mean MSE ratios under strict leave-one-city-out dynamics validation.

6.6 Auxiliary sampled rollout

The sampled BusSimEnv rollout is not primary evidence, but it checks whether the learned one-step action preferences are grossly inconsistent with executable line environments. Under a fair seeded rollout over ten Austin/Halifax line episodes, weighted CFCMTreaches parity with the dense policy: mean reward is -1121.574 for weighted CFCMTand -1121.612 for H2O^{+}. The headway-error metric is slightly worse for weighted CFCMT, 588.2 seconds versus 585.6 seconds. Thus rollout supports a limited claim: after schema and reward alignment, CFCMTdoes not collapse in executable simulation, but the closed-loop advantage remains too small to serve as the main evidence.

7 Discussion

Why factored residuals help. The dense residual baseline has enough capacity to fit cross terms in source data, but the same flexibility can mix city-specific demand, speed, and headway

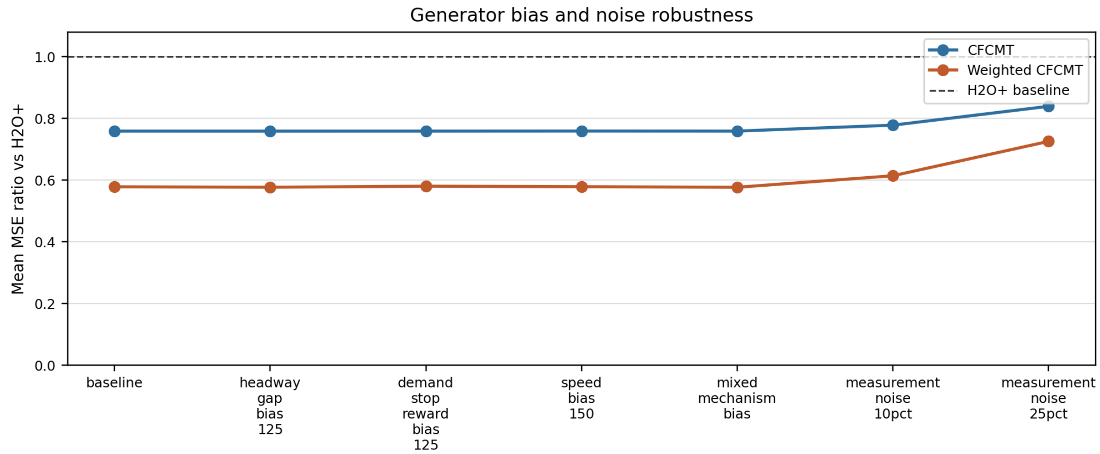

Figure 6: Generator-bias and measurement-noise robustness under strict leave-one-city-out evaluation.

effects. CFCMTrestricts each output to a smaller mechanism-specific parent set. This constraint appears helpful in the strict cross-city setting: it reduces total MSE on all four targets and makes source weighting more effective.

What the source weighting does and does not show. Similarity weighting improves the mean strict leave-one-out ratio from 0.759 to 0.578 and wins over unweighted CFCMTon three of four targets. The Singapore target is the exception, where weighting is slightly worse than unweighted CFCMT. This suggests that static similarity is useful but imperfect; city descriptors built from open data may miss route-level operational differences that matter for transfer.

Policy evidence remains secondary. The one-step policy validation and sampled BusSimEnv rollout are useful checks, but they do not replace a closed-loop operational benchmark. The rollout result reaches reward parity with $H2O^{+}$only after aligning BusSimEnv observations to the static feature schema and using identical line-level random seeds across policies. That correction is necessary for a fair comparison, but the resulting rollout margin is small. The strongest claim should therefore remain dynamics transfer plus policy-proxy improvement, not full policy dominance.

Static-derived validation is not operational validation. The central limitation is target construction. The experiments use full-route open static data and deterministic transition rules to create paired simulator/target transitions. This is valuable for reproducibility and reviewer scrutiny, but it is not the same as evaluating on AVL/APC trajectories or a calibrated live SUMO rollout. The paper should therefore avoid claims such as “CFCMT is better on real bus systems”. A precise claim is: under open static-data-derived cross-city validation with little or no target-city calibration, mechanism-factored residual transfer generalizes better than dense residual correction.

How to strengthen the next version. The most valuable addition would be one city with vehicle-level AVL/APC trajectories, even if only for a subset of routes. The second priority is to expand executable line environments so sampled rollout can move from auxiliary sanity check to a stronger policy benchmark. The third priority is to add more cities, because city-level generalization is ultimately limited by the number of held-out cities.

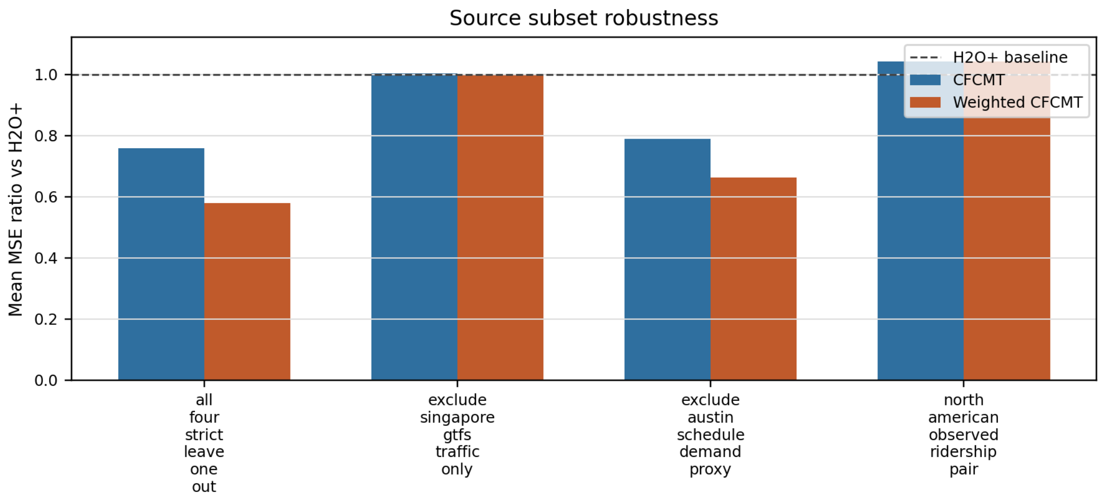

Figure 7: Source-subset robustness. Removing data-source families reveals where the claim becomes weak.

8 Conclusion

We presented CFCMT, a causal-factored residual transfer approach for bus holding control under cross-city open-transit validation. Compared with a dense H2O$^+$-style residual baseline, CFCMTimproves strict leave-one-city-out dynamics on all four target cities, and source-city similarity weighting further improves the mean MSE ratio. One-step policy validation also favors CFCMTon most reward/regret and headway metrics and on all bunching-rate comparisons. At the same time, subset robustness shows that the advantage depends on source diversity, and generator robustness shows that not every perturbation preserves all-target wins. The resulting paper claim is intentionally bounded: CFCMTis a promising mechanism-factored alternative to dense residual correction for low-calibration cross-city transfer, but operational deployment claims require real trajectory validation.

A Additional Tables

A.1 Target-construction audit

city	demand_evidence	traffic_evidence	lines	transitions	uncal_mse	line_mse_p25	line_mse_p50	line_mse_p75
Singapore	observed stop OD/PV monthly	observed speed bands	789	1243488	2256.5339	220.173	324.7508	541.4336
Austin / CapMetro	schedule proxy	schedule derived	234	310272	1301.3027	242.2482	615.8957	3043.8061
Halifax Transit	route APC apportioned	schedule derived	176	266448	1111.9223	199.2164	620.2908	1614.3016
MBTA	stop board/alight apportioned	schedule derived	940	969360	1216.3716	180.9625	459.6093	1040.0624

Table 5: Target-construction audit and route-level nontriviality check.

budget	splits	mean_no_cal_ratio	mean_cal_ratio	no_cal_wins	cal_wins	mean_calib_lines	oracle
0.0	4	0.5782	0.5782	4	4	0.0	False
0.01	4	0.5807	0.5801	3	3	3.5	False
0.05	4	0.5863	0.5831	3	3	23.0	False
0.1	4	0.6165	0.6094	3	3	50.5	False
0.25	4	0.6772	0.6596	3	3	132.75	False
1.0	4	0.7872	0.7242	3	3	534.75	True

Table 6: Target-route calibration-budget summary. Budget 1.0 is an in-sample oracle reference.

mechanism	targets	weighted_wins	mean_cfcmt_ratio	mean_weighted_ratio
demand_stop_reward	4	4	0.8197	0.7706
headway_gap	4	3	0.7586	0.5743
speed	4	4	0.8203	0.6684

Table 7: Per-mechanism strict leave-one-city-out error ratios.

A.2 Calibration budget

A.3 Per-mechanism errors

A.4 Mechanism and capacity ablations

A.5 Route-bootstrap confidence intervals

A.6 Configured six-split dynamics validation

A.7 Source-weighting sensitivity

A.8 Source-size sensitivity and single-source transfer

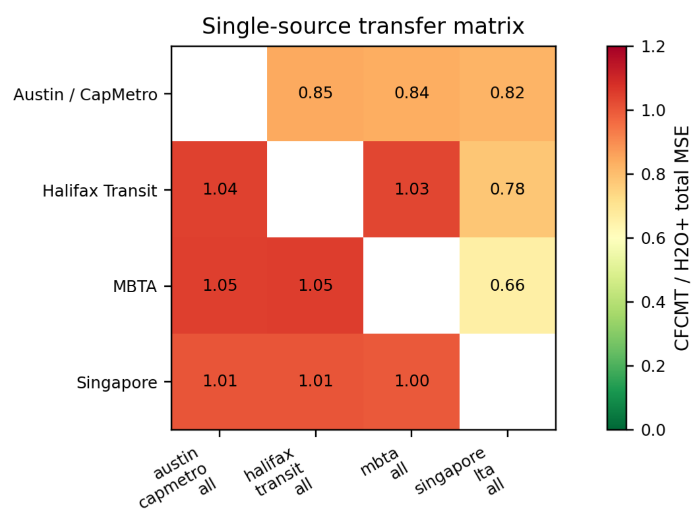

Figure 8: Single-source transfer matrix. Cells report CFCMT/H2O total-MSE ratio.

family	wins_vs_h2oplus	mean_ratio_vs_h2oplus
cfcmt_shared_sparse	4	0.7502
cfcmt_no_action_interaction	4	0.7583
cfcmt_full	4	0.7587
dense_matched_sparse	3	0.7716
dense_all_parents_no_pairwise	3	0.9332
cfcmt_action_time_only	1	7.3401
cfcmt_random_mechanism_grouping	1	7.5744

Table 8: Mechanism and capacity ablations. Lower mean ratios indicate better transfer relative to $H2O^+$.

target_city	lines	cfcmt_ratio	cfcmt_ratio_ci95	weighted_ratio	weighted_ratio_ci95
Singapore	789	0.9964	[0.991, 1.000]	0.9986	[0.993, 1.002]
Austin / CapMetro	234	0.7375	[0.670, 0.817]	0.3985	[0.354, 0.447]
Halifax Transit	176	0.7166	[0.529, 0.871]	0.5638	[0.310, 0.783]
MBTA	940	0.5844	[0.520, 0.643]	0.352	[0.257, 0.438]

Table 9: Route-level bootstrap confidence intervals under strict leave-one-city-out dynamics validation.

A.9 Generator robustness

A.10 Source-subset robustness

A.11 Auxiliary sampled rollout

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split	target_city	train_transitions	target_transitions	h2oplus_total_mse	cfcmt_total_mse	cfcmt_vs_h2oplus_ratio	weighted_cfcmt_vs_h2oplus_ratio	weighted_cfcmt_vs_unweighted_ratio
source_open_na_to_singapore	Singapore	1546080	1243488	2254.8548	2246.6693	0.9964	0.9986	1.0022
source_singapore_austin_halifax_to_mbta	MBTA	1820208	969360	318.9925	186.4177	0.5844	0.352	0.6022
leave_one_city_out_all::singapore_lta_all	Singapore	1546080	1243488	2254.8548	2246.6693	0.9964	0.9986	1.0022
leave_one_city_out_all::austin_capmetro_all	Austin / CapMetro	2479296	310272	91.9724	67.8254	0.7375	0.3985	0.5404
leave_one_city_out_all::halifax_transit_all	Halifax Transit	2523120	266448	97.1175	69.5992	0.7166	0.5638	0.7867
leave_one_city_out_all::mbta_all	MBTA	1820208	969360	318.9925	186.4177	0.5844	0.352	0.6022

Table 10: Configured cross-city dynamics splits. This table includes repeated target cities and is treated as appendix evidence rather than the primary independent-sample result.

temperature	floor	wins_vs_h2oplus	wins_vs_unweighted_cfcmt	mean_ratio_vs_h2oplus	mean_ratio_vs_unweighted_cfcmt
0.5	0.0	6	4	0.6025	0.7449
0.5	0.05	6	4	0.6033	0.746
0.5	0.1	6	4	0.6041	0.747
1.0	0.0	6	4	0.6101	0.7554
1.0	0.05	6	4	0.6106	0.756
1.0	0.1	6	4	0.611	0.7566
2.0	0.0	6	4	0.6145	0.7614
2.0	0.05	6	4	0.6147	0.7617
2.0	0.1	6	4	0.6149	0.762

Table 11: Source-weighting sensitivity over temperature and floor.

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source_size	splits	cfcmt_wins_vs_h2oplus	mean_ratio_vs_h2oplus
1	12	6	0.9272
2	12	8	0.8705
3	4	4	0.7587

Table 12: Mean cross-city ratio by number of source cities.

scenario	splits	weighted_wins_vs_h2oplus	unweighted_wins_vs_h2oplus	weighted_wins_vs_unweighted	mean_unweighted_ratio_vs_h2oplus	mean_weighted_ratio_vs_h2oplus	mean_weighted_ratio_vs_unweighted
baseline	4	4	4	3	0.7587	0.5782	0.7329
headway_gap_bias_125	4	3	3	3	0.7587	0.5768	0.7301
demand_stop_reward_bias_125	4	4	4	3	0.7587	0.5802	0.737
speed_bias_150	4	4	4	3	0.7589	0.5786	0.7332
mixed_mechanism_bias	4	3	3	3	0.7588	0.5767	0.7295
measurement_noise_10pct	4	4	4	3	0.778	0.614	0.7648
measurement_noise_25pct	4	4	4	3	0.8391	0.7255	0.851

Table 13: Generator-bias and measurement-noise robustness.

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subset	cities	splits	weighted_wins_vs_h2oplus	unweighted_wins_vs_h2oplus	mean_unweighted_ratio_vs_h2oplus	mean_weighted_ratio_vs_h2oplus	mean_weighted_ratio_vs_unweighted
all_four_strict_leave_one_out	Singapore, Austin / CapMetro, Halifax Transit, MBTA	4	4	4	0.7587	0.5782	0.7329
exclude_singapore_gtfs_traffic_only	Austin / CapMetro, Halifax Transit, MBTA	3	1	1	1.0025	0.9984	0.9964
exclude_austin_schedule_demand_proxy	Singapore, Halifax Transit, MBTA	3	3	3	0.7898	0.6626	0.8124
north_american_observed_ridership_pair	Halifax Transit, MBTA	2	0	0	1.0421	1.0421	1.0

Table 14: Source-subset robustness.

policy	episodes	mean_reward	mean_headway_abs_error	mean_hold_seconds
no_hold	10	-1138.411	601.9113	0.0
fixed_30	10	-1151.2901	598.4459	30.0
h2oplus_dense_policy	10	-1121.612	585.5526	3.117
cfcmt_mechanism_policy	10	-1122.2105	589.1995	3.4897
cfcmt_similarity_weighted_policy	10	-1121.574	588.2357	3.2859

Table 15: Auxiliary sampled BusSimEnv rollout. This result is not primary evidence because only a small number of runnable line environments are available.